

RAJ DESAI

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SUMMARY

Software Engineer who ships distributed systems and AI-powered tools, from an 8-service contract-review platform with saga orchestration to async code-review pipelines under hard latency limits. Strong in API design, real-time systems, and LLM integration across Python and TypeScript. MS in Computer Science from UT Dallas.

EDUCATION

The University of Texas at Dallas | MS – Computer Science

Dallas, TX | May 2026

Relevant Courses: Design and Analysis of Algorithms, Machine Learning, NLP, AI, Big Data Analytics, Database Design.

University of Mumbai | BE – Computer Engineering

Mumbai, India | May 2024

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, Dart, C/C++, HTML, CSS

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Server-Sent Events, Celery, RabbitMQ, BullMQ, Redis, OAuth 2.0, JWT

Data: PostgreSQL, pgvector, MySQL, MongoDB, Prisma ORM, SQLAlchemy, Pandas, NumPy

AI & LLM: LLM APIs (OpenAI, Gemini, Anthropic), LiteLLM, model routing & fallback, semantic caching, RAG, FAISS, Sentence Transformers

Frontend: React, Next.js, Tailwind CSS, Framer Motion, Recharts

Cloud & DevOps: AWS (Lambda, CloudWatch, DynamoDB, SNS), Docker, Docker Compose, GitHub Actions, CI/CD, OpenTelemetry, Vercel, Railway, Git

PROFESSIONAL EXPERIENCE

Software Engineer Intern - Hydracoral Technologies

Jan 2023 – Dec 2023 | Mumbai, India

- Integrated third-party facial-recognition APIs into a Flutter/Dart onboarding flow, cutting user verification time by 30% in staging
- Reduced app crash rate by 20% by diagnosing and resolving 15+ bugs across state management, API error handling, and memory leaks, using Firebase Crashlytics and structured debugging workflows
- Lifted usability scores by 20% by building responsive cross-platform UI components, validated through structured user-testing
- Owned CI/CD pipelines (Git, Bitbucket) for a 3-person Agile team and context-switched daily across frontend, backend, and DevOps, integrating REST APIs for JWT auth, data sync, and release automation

PROJECTS

ClauseGuard - AI-Powered Contract Review Platform

Apr 2026 – May 2026

- Designed and deployed an 8-service distributed system (FastAPI, Celery, PostgreSQL/pgvector, Redis, RabbitMQ, MinIO) orchestrated via Docker Compose with health checks, graceful degradation, and OpenTelemetry distributed tracing
- Engineered a saga-pattern pipeline across Celery and RabbitMQ with circuit breakers on LLM calls, fan-out parallelism for clause classification, and Redis Pub/Sub SSE for live frontend progress
- Reduced LLM costs by 30% via semantic caching with pgvector cosine similarity, structured JSON outputs via Pydantic, and model-tier routing through a pluggable LiteLLM fallback layer

Yoru - Cinematic Ambient Music Streaming Web App

Jan 2026 – Feb 2026

- Shipped a cinematic streaming app with Next.js 15, TypeScript (strict), and Tailwind CSS, orchestrating 21 scenes with Framer Motion crossfades and dual-channel Howler.js audio mixing
- Engineered custom React hooks (useKeyboard, useIdleDetection, useAudio) that encapsulate reusable interaction logic with WCAG 2.1 AA keyboard navigation, ARIA support, and screen-reader compatibility
- Maintained >80% test coverage with Vitest and React Testing Library, enforcing component-driven architecture with strict TypeScript and isolated unit tests for each custom hook

ZeroWall - Serverless Zero-Trust API Gateway on AWS (3-person course project)

Spring 2026

- Built the observability and monitoring layer for a 7-service AWS serverless zero-trust API gateway: CloudWatch log groups, custom metrics, and alarms, plus SNS email alerting
- Wrote a scheduled anomaly-detection Lambda (Python) that scans the DynamoDB audit log every 5 minutes for brute-force and credential-stuffing patterns, plus a Postman test suite for the gateway's endpoints

PR Sensei - AI-Powered GitHub Code Review App

Nov 2025 – Jan 2026

- Solved GitHub's hard 10-second webhook timeout against 5 to 15-second LLM calls by engineering an async pipeline with Redis and BullMQ that handles retries, rate limiting, and graceful degradation under bursty load
- Shipped an AI GitHub App that posts PR summaries and inline comments in under 30 seconds via a pluggable OpenAI/Gemini provider layer triggered by webhook events
- Designed a multi-tenant PostgreSQL schema (Prisma ORM) feeding a React + Recharts dashboard for review analytics, file-hotspot tracking, and per-repository filtering